Fiscal Year:	FY 2011	**Task Last Updated:**	FY 01/11/2011
PI Name:	Willey, Jeffrey S. Ph.D.		
Project Title:	Radiation Effects on Bone Tissue and Cells in Reduced Gravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27157-0001	**Congressional District:**	5
Comments:	NOTE: PI formerly at Clemson University when NSBRI Postdoctoral Fellow Feb 2008-Oct 2010 (Ed., 12/18/2014)		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-02 Postdoctoral Fellowships
Start Date:	02/01/2008	**End Date:**	10/01/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/01/2010 (from 9/30/2010) to accommodate final reporting (Ed. Jan 2011) NOTE: End date changed to 09/30/2010 (from 1/31/2011) per NSBRI (8/2010) NOTE: End date changed to 1/31/2011 (from 1/31/2010) per NSBRI (1/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bateman, Ted (MENTOR/Clemson University)		
Grant/Contract No.:	NCC 9-58-PF01403		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	POSTDOCTORAL FELLOWSHIP The spaceflight environment presents numerous challenges to astronauts' bone health, including the bone-debilitating effects of reduced gravity. A reduction in bone volume and strength among astronauts represents a serious, mission-critical problem. Preliminary data and published accounts indicate that exposure to doses as low as 1 Gy of multiple types of ionizing radiation results in trabecular bone loss. However, it is unclear how bone already compromised by microgravity or partial gravity will be affected by exposure to space radiation. Our intent is to enhance the understanding of the effects of radiation on bone health, particularly within a reduced gravity environment. Our ultimate goal is to develop appropriate countermeasures for bone loss in astronauts during exploratory missions. Specific Aim 1: Examine the response of bone to a combination of spaceflight-associated challenges, including radiation exposure together with reduced limb bone loading. Approach Aim 1: Skeletally mature female C57BL/6 mice were exposed to a 1 Gy dose of protons at Loma Linda Medical Center followed by a 4-week period of hindlimb suspension. Main Findings Aim 1: 1) The combination of radiation and hindlimb suspension resulted in greater bone loss and deterioration of trabecular microarchitecture than the two challenges individually. 2) Osteoclasts appear to be more active as an early response in both irradiated groups (irradiated only and irradiated + suspension) than non-irradiated groups. Specific Aim 2: Examine mechanistic causes of bone and cartilage loss (at the molecular, cellular, and tissue level) after radiation and unloading on rodent hindlimbs. Approach Aim 2: Year 1: 13-week old C57BL/6 mice received whole body radiation with 2 Gy X-rays. X rays were chosen for these initial studies, as bone loss from a 1 Gy dose has been shown to be similar for both types at one month (preliminary data). Year 2 and 3: In an attempt to expand this NSBRI-funded technology to develop countermeasures for spaceflight and clinical applications, 16-week old mice were exposed to a 2 Gy dose of X rays. Some irradiated animals were given a commercially available osteoporosis medication (risedronate) to try to prevent any changes. Bone was examined at weeks 1, 2, and 3. Additionally, the right hindlimbs of twenty-week old Sprague-Dawley rats received a dose of radiation modeling what the human hip receives during cancer radiotherapy. Risedronate was applied to try to prevent these changes. Year 3: The effects of radiation on astronaut joint cartilage is completely unknown. Mouse limbs were irradiated with a 2 Gy dose of X-rays to identify is strength changes could occur in the cartilage. Pig chondrocytes received 10 Gy of gamma rays in an attempt to identify mechanisms for radiation-induced erosion. Main Findings Aim 2: 1) Radiation increased osteoclast number and activity as an early, acute response to radiation leading to early bone loss within the first week. Applying the osteoporosis medication completely prevented loss of bone. 2) By applying a dose of radiation that precisely models what a human hip receives during treatment for a pelvic tumor to a rat model, bone deterioration occurs within 2 weeks. Strength of the bone was reduced as well. These changes were prevented with the osteoporosis medication.		

	3) Stiffness of cartilage is significantly reduced by only 1 week after exposure to radiation, which is a novel finding. 4) Irradiation of pig cartilage results in the production of proteins (MMP's and aggrecanases) by 1 day after treatment that can cause early loss of cartilage. Application of antioxidants (MnTBAP) prevents this effect, and thus may serve as a potential countermeasure for joint damage in space. Future studies must examine reduced loading and radiation on total joint health (bone and cartilage) with countermeasures (e.g., antiresorptive agents; antioxidants).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The following Aim for this project supported by NSBRI has direct clinical relevance. The notion that radiation can cause an active increase in osteoclast activity and lead to both loss of bone volume, architecture, and strength is a truly novel concept funded from NSBRI. This early loss of bone occurs after modeled spaceflight scenarios (e.g., 1 Gy of proton exposure), and if we exactly model the dose absorbed by the human hip after radiation-therapy for a pelvic tumor. The deterioration of bone is correlated with loss of strength. We prevented all of these early changes by applying an agent (the bisphosphonate risedronate) that targets the osteoclast and decreases its function. Thus, we can potentially develop countermeasures to prevent radiation-induced loss of strength among not only astronauts, but also among cancer survivors. Radiation-induced bone damage is a concern among survivors of pelvic tumors. Improvements in cancer treatment and diagnosis have led to increased long-term cancer survivorship. However, this growing population of long-term survivors are at risk of developing deleterious effects in normal tissues resulting from the use of therapeutic radiation. Fractures of the hip are certainly noted among survivors of pelvic cancers, and only where the bone absorbs radiation. Likewise, vertebral fractures are observed at sites that absorb dose during lung cancer treatment. Fractures of the hip are a significant cause of morbidity and mortality among elderly patients: 85% of post-menopausal women who fracture a hip have impaired mobility for life, and ~20% will die within a year of the fracture. Thus, if we can prevent fractures among cancer survivors, particularly of the hip, then we can substantially improve the quality of lives of these patients. We intend on targeting how radiation can influence the osteoclast (and the cells and microenvironment of the whole joint) to try to prevent fractures and joint erosion among cancer survivors.
Task Progress:	Substantial progress was made during the course of the past year in terms of identifying that loss of bone strength occurs after irradiation. Importantly, this loss of bone strength is observed in rodent models after receiving a course of radiation treatment that models the dose a human hip receives during pelvic cancer treatment, and we can prevent these changes by applying countermeasures. Specific Aim 2: Examine the cause of radiation-induced loss of strength and test preventative countermeasures (antiresorptive; anti-inflammatory) to prevent any early response that can lead to increased osteoclast activity. Progress: Right hind limbs of rats were exposed to radiation using a LINAC, which is used to apply therapeutic radiation to human cancer patients. Over a 2-week period, right hind limbs were exposed to a total dose of 16 Gy of 6 MV X-rays delivered in 4 fractions using the LINAC. This treatment regime models the dose of radiation absorbed by the hips of women treated for cervical cancer. Risedronate (a commercially available osteoporosis treatment) was applied in an attempt to prevent loss of strength. Bone strength was determined from finite element analysis of microCT images at 2, 4, and 6 weeks. The estimated strength of the trabecular bone is significantly weaker by 4 weeks after exposure, therefore loss of strength occurs with loss of bone. Again, loss of bone strength was prevented with the antiresorptive osteoporosis medication. Specific Aim 2: Identify if radiation can deteriorate joint cartilage, and apply preventative countermeasures. Progress: Articular (joint) cartilage of mice after exposure to ionizing radiation appears to deteriorate. C57BL/6 female mice were given a 2 Gy whole-body dose of X rays, which we have previously shown to cause bone loss similar to low dose proton exposures relevant to the spaceflight environment. The articular cartilage on the distal end of the femur was the site used for mechanical testing and histological analysis. The irradiated articular cartilage is much less stiff than non-irradiated cartilage. Therefore, radiation could have detrimental effects on the function of articular cartilage. Additionally, we attempted to determine if irradiated cartilage can produce proteins (MMP-3, ADAMTS4 and ADAMTS5) capable of degrading itself, which can happens in states like osteoarthritis. Pig chondrocytes were irradiated with a 10 Gy dose of gamma rays. One day after exposure, levels of MMP-3 and ADAMTS4 and 5 were increased in the cells media. These changes were prevented by applying MnTBAP (an antioxidant) to the cells immediately before irradiating. Therefore, radiation appears to induce oxidative stress, which may cause early loss of cartilage. Oxidative stress has also been shown to activate osteoclasts and stimulate bone loss. Thus, future studies may want to examine the effects of radiation-induced oxidative stress on whole joint health (bone and cartilage).
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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